A Decision Support System for Startup Viability Based on Market Trends, Saturation, and Business Dynamics*

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Abstract

Startup businesses in the Philippines often struggle to identify viable ventures due to reliance on guesswork, peer advice, and trends, leading to poor product–market fit, oversaturation, and high failure rates. The lack of accessible, data-driven tools further limits informed decision-making. This study presents the design of a localized business viability assessment system integrating business directories, online reviews, social media engagement, and geospatial data. It evaluates three dimensions: popularity, volatility, and competition. Popularity reflects customer feedback, volatility captures business openings and closures, and competition measures business density. Results show the Volatility Model (Logistic Regression) achieved the highest performance (85.23% ROC-AUC), with *has_recent_renewal* as the strongest predictor. Model optimization improved performance to 85.12% accuracy and 91.18% ROC-AUC. Although the 90% accuracy target was not met due to dataset and prediction constraints, results align with literature benchmarks. Restaurants exhibit the highest risk, while Hotels/Resorts show the strongest viability.

1 Introduction

Research on SME digital transformation indicates that a large proportion of SMEs fail to survive beyond their early years due to challenges in market adaptation, competition, and lack of strategic guidance, with survival rates often below 50% after five years [1]. For the failure driven aspect linked to poor planning and reliance on limited information (such as anecdotal advice or trends), while direct Philippine studies linking these specific behaviors to failure are limited in formal literature, broader SME research identifies lack of market research and inadequate information use as major contributors to poor business outcomes globally factors that are widely recognized in entrepreneurship scholarship as leading causes of early business failure. In the Philippines, micro, small, and medium enterprises represent over 99 % of registered business establishments, yet many of these firms struggle to grow and succeed due to inadequate structured market research, planning, and strategic decision making factors that are widely associated with higher small business failure rates.[2]

Recent advances in machine learning and cloud based analytics have significantly enhanced predictive, data driven tools for business decision making, enabling scalable analytics and real time insights that support more informed strategic planning.[3]. By analyzing market trends, competitor density, and business volatility, such systems can provide actionable insights previously inaccessible to small-scale entrepreneurs.

Recent studies highlight the role of online reviews and competition indicators in predictive business analytics: comprehensive surveys show that machine learning models incorporating market features and competitive signals can effectively model business success [4], dynamic review based frameworks derive competitive intelligence from temporal review data [5], and empirical work demonstrates that online reputation metrics significantly relate to small business performance in competitive markets [6].

Entrepreneurs often lack access to structured, data-driven guidance, which contributes to poor venture selection and high business failure rates. To address this challenge, the study focuses on collecting

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and curating local business data, developing a machine learning model using Logistic Regression and K-Nearest Neighbors to predict business viability based on popularity, competition, and market volatility, and validating the model's predictive accuracy to ensure practical applicability for entrepreneurs and policymakers. The system targets SMEs in core business sectors with publicly available online data while excluding large franchises, internal operational metrics, and pricing strategies. Findings are constrained to structured, accessible datasets and may not be generalizable to other regions or market contexts.

2 Design and Methodology

The methodology and system architecture for assessing SME business viability across areas in Bohol Province using a data-driven approach. The system evaluates three key factors popularity, competition, and market volatility through dedicated model branches. Each branch generates a calibrated probability, which is combined using learned weights to produce a final feasibility score. This chapter will explain the data collection procedures, model selection, preprocessing techniques, validation methods, system design, evaluation protocols, and a comparative analysis of the ensemble model against alternative modeling approaches.

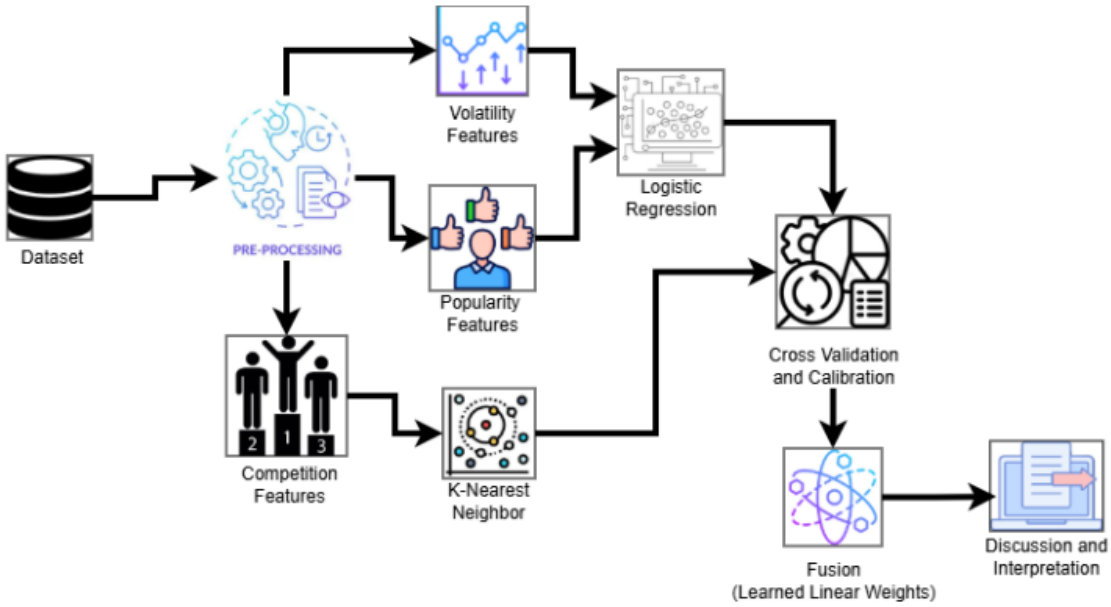


Figure 1: Conceptual Framework

The proposed system uses a multi-branch ensemble to evaluate business viability based on popularity, competition, and market volatility. Data are sourced from online review platforms for popularity metrics, geographic data for competition context, and local government records for historical openings, closures, and business lifespans. The dataset undergoes preprocessing and feature engineering, including de-duplication, handling missing values, and standardizing categories and locations. Three distinct feature sets are constructed: popularity (ratings, reviews, social engagement, nearby attractions) [7], competition (competitor counts, distances, clustering, operating hours)[8], and volatility (business openings/closures, average lifespan). Skewed and distance-based features are appropriately transformed or standardized to improve model interpretability and performance.

1. **Parallel Branch Modeling.**
   Each dimension is modeled with an algorithm suited to its data characteristics:

   - **Popularity branch – Logistic Regression.** A regularized Logistic Regression (LR) model uses popularity-related features to estimate a calibrated feasibility probability, P_{pop},

based on the business's online presence.

- **Competition branch – K-Nearest Neighbors.** A distance-weighted KNN model within the same business category captures local saturation and proximity patterns, producing a calibrated competition probability, P_{comp}.
- **Volatility branch – Logistic Regression.** A second Logistic Regression model utilizes historical opening, closure, and business lifespan statistics to estimate the volatility probability, P_{vol}.

The outputs of these three branches are subsequently integrated to generate a composite viability score for each business.

2. **Calibration and Validation.**

   The system ensures reliable and comparable branch outputs through stratified k-fold cross-validation with per-branch probability calibration (Platt scaling or isotonic regression). Preprocessing artifacts and model parameters are fitted only on the training folds to prevent information leakage. Feasibility labels are defined according to business outcomes within a specified observation period.

   The calibrated probabilities are combined using a learned linear fusion to produce the final feasibility probability:

$$P_{\text{final}} = w_{\text{pop}} P_{\text{pop}} + w_{\text{comp}} P_{\text{comp}} + w_{\text{vol}} P_{\text{vol}} + b, \tag{1}$$

   where w_{pop}, w_{comp}, and w_{vol} are the learned fusion weights, and b is the bias term.

   The fusion weights are optimized through cross-validated hyperparameter searches to maximize evaluation metrics such as F1-score or PR-AUC. A tri-band decision policy is then applied using two thresholds learned during validation.

   The system outputs the final viability probability, P_{final}, together with the three branch probabilities (P_{pop}, P_{comp}, and P_{vol}) to provide interpretable predictions by exposing the contribution of each dimension. After model selection, all preprocessing pipelines, branch models, calibration mappings, and fusion weights are fixed for deployment, with periodic retraining performed as new business data become available.

3. **Classification Scheme.**

   The final probability is mapped into a three-level feasibility classification that provides actionable recommendations:

   - **Feasible** ($P_{\text{final}} \geq 0.60$): Strong indicators of sustained business operation.
   - **Moderate Risk** ($0.40 \leq P_{\text{final}} < 0.60$): Mixed indicators requiring additional market analysis and due diligence.
   - **Not Feasible** ($P_{\text{final}} < 0.40$): High-risk indicators suggesting market avoidance or substantial business strategy revision.

3 Research Procedure.

The proposed methodology follows a structured data-to-model workflow to produce reliable and locally relevant predictions of business feasibility. Raw data are collected from multiple sources and integrated using a canonical *business_id* to ensure consistency across platforms. The collected information includes business profiles, popularity metrics (e.g., ratings and reviews from Google and Facebook), business hours, registration status, competitor information, and nearby points of interest (POIs). Particular attention is given to the 2019–2021 COVID-19 period to capture its impact on business performance. A mapping process is applied to ensure that records from different platforms refer to the same business. Although businesses operating between 2015 and 2025 are considered for feature construction, only those satisfying the defined eligibility criteria are used for model training and evaluation.

The dataset contains approximately 170–220 businesses from four categories: hotels and resorts, restaurants, salons and spas, and cafés. Each business is represented by approximately 15–20 attributes covering identification, location, popularity, competition, and operational characteristics. After data cleaning and feature construction, the dataset is divided temporally, with earlier business records used for training and more recent records used for testing. Businesses without sufficient observation periods are excluded from training and testing but are retained for feature extraction.

1. **Raw Data Requirements (Pre-processing)**
   Prior to feature engineering, the system collects several relational tables linked through the canonical *business_id*. The raw attributes include:

   - **Canonical business information:** business_id, official_name, category (Hotel/Resort, Restaurant, Salon/Spa, Café), address, barangay, latitude, longitude, and data_snapshot_date.
   - **Popularity sources:** platform, platform_business_id, platform_url, rating, rating_scale_max, review_count, review recency, followers/likes/check-ins, and collected_at.
   - **Business hours:** weekly operating schedule and time zone.
   - **Business lifecycle:** permit number, opening date, closure date (if applicable), business status, and last registry update.
   - **Competitor information:** business identifier, category, geographical coordinates, barangay, opening date, basic popularity indicators, and data snapshot.
   - **Points of Interest (POIs):** POI type, name, coordinates, data source, and collection timestamp.
   - **Platform mapping:** platform, platform business identifier, canonical business identifier, matching confidence, and matching method.

2. **Data Collection**
   Multiple data sources are utilized according to the information required by each predictive branch.

   - **Popularity data**
     - Sources: Google Maps/Places, Facebook Business Pages, and, where available, Booking, Agoda, and TripAdvisor.
     - Metrics: average rating, review count, review recency, social-media presence, number of active platforms, and nearby attraction points.
   - **Competition data**
     - Sources: Google Maps APIs, OpenStreetMap, and barangay-level directories.
     - Metrics: number of competitors within 1 km, nearest competitor distance, city-wide competitor count, business clustering, average competitor age, and business-hour overlap.
   - **Volatility data**
     - Source: Tagbilaran City Hall through official records.
     - Metrics: business type, number of openings, number of closures, and business lifespan.

3. **Data Splitting Strategy**
   A temporal data splitting strategy is adopted to avoid information leakage. Business records from 2015–2019 are used for model training, representing stable pre-pandemic market conditions. Records from 2022 onward are reserved for testing to evaluate post-pandemic generalization performance. Data from 2020–2021 are intentionally excluded because COVID-19 introduced abnormal market conditions that could bias the learning process.

4. **Logical Design**

   The proposed decision-support system evaluates business feasibility using three complementary dimensions: popularity, competition, and volatility. Each dimension is modeled independently and produces a calibrated probability score.

   The branch probabilities are defined as

$$P_{\text{pop}} = LR_{\text{pop}}(x_{\text{pop}}), \tag{2}$$

$$P_{\text{comp}} = KNN_{\text{comp}}(x_{\text{comp}}), \tag{3}$$

$$P_{\text{vol}} = LR_{\text{vol}}(x_{\text{vol}}). \tag{4}$$

   The calibrated probabilities are then combined using a learned linear fusion model,

$$P_{\text{final}} = w_{\text{pop}}P_{\text{pop}} + w_{\text{comp}}P_{\text{comp}} + w_{\text{vol}}P_{\text{vol}}, \tag{5}$$

   where the fusion weights are learned during model training.

   The system reports both the final feasibility probability and the three individual branch probabilities to improve model interpretability. If one branch contains missing information, the remaining weights are re-normalized and the prediction is flagged with a higher uncertainty level.

5. **Benchmark Expectations**

   Model performance will be evaluated against recent business viability prediction studies. Following the literature, F1-score is expected to fall within the reported range of 76%–94%, with an aspirational target of approximately 90% accuracy. In addition to accuracy and F1-score, PR-AUC and Brier Score are employed to evaluate discrimination capability and probability calibration.

4 Result and Discussion

4.1 Dataset Composition and Preprocessing

The data extraction pipeline effectively integrated business information from multiple sources, including the Province of Bohol Business Permits and Licensing Office records, Google Places API, and other online platforms. The resulting dataset, as summarized in Table 1, showed a moderate class imbalance, with about 65% of businesses categorized as Feasible. Recognizing this imbalance, the evaluation prioritized robust performance metricssuch as precision, recall, F1-score, and ROC-AUCrather than relying solely on accuracy, to ensure a more reliable assessment of the model's performance.

Table 1: Dataset Composition Summary

Attribute	**COVID-19 Included**	**COVID-19 Excluded**
Total Businesses	716	628
Training Set	501 (2017-2021)	413 (2017-2019)
Test Set	215 (2022-2025)	215 (2022-2025)
Feasible (Label = 1)	462 (64.5%)	412 (65.6%)
Not Feasible (Label = 0)	254 (35.5%)	216 (34.4%)

4.2 Baseline Model Performance

The Volatility Model, implemented using Logistic Regression, achieved the highest standalone performance, attaining an ROC-AUC of 85.23% and an F1 score of 89.44% as shown in Table 2, thereby underscoring the critical role of operational stability indicators in predicting business viability.

Table 2: Baseline Model Performance Summary

Model	Algorithm	Accuracy	F1 Score	ROC-AUC
Popularity	Logistic Regression	74.91%	84.09%	58.84%
Competition	KNN	63.44%	72.13%	69.80%
Volatility	Logistic Regression	83.15%	89.44%	85.23%

4.3 Category-Specific Analysis

Table 3 presents feasibility and risk classifications across four business categories, revealing notable variations in viability and associated risk levels. For Hotels/Resorts ($n = 31$), a majority are classified as feasible (58.1%), followed by moderate risk (35.5%), and a small proportion deemed not feasible (6.5%), resulting in the lowest risk rate among all categories. In contrast, Cafés ($n = 141$) exhibit a very low feasibility rate (4.3%), with most falling under moderate risk (82.3%) and 13.5% categorized as not feasible, indicating relatively higher uncertainty.

Table 3: Tri-band Classification by Category

Category	Total	Feasible	Moderate Risk	Not Feasible	Risk Rate
Hotel/Resort	31	18 (58.1%)	11 (35.5%)	2 (6.5%)	6.5%
Cafe	141	6 (4.3%)	116 (82.3%)	19 (13.5%)	13.5%
Salon/Spa	162	31 (19.1%)	109 (67.3%)	22 (13.6%)	13.6%
Restaurant	294	50 (17.0%)	152 (51.7%)	92 (31.3%)	31.3%

4.4 Fused Model Performance Comparison

The comparison highlights a clear trade-off across performance metrics. Although the model that includes COVID-19 data attained higher accuracy (79.93% vs. 73.48%) and recall (90.95% vs. 74.76%), the model excluding COVID-19 data achieved better precision (+4.43%) and ROC-AUC (+1.08%). Based on Table 4, the increase in precision suggests a reduction in false positives, indicating that the model adopts a more conservative approach and is more reliable in correctly identifying genuinely viable businesses.

Table 4: Weighted Fusion Model Performance - COVID-19 Comparison

Metric	Covid Included	Covid Excluded	Difference
Accuracy	79.93%	73.48%	-6.45%
Precision	83.77%	88.20%	+4.43%
Recall	90.95%	74.76%	-16.19%
F1 Score	87.21%	80.93%	-6.28%

4.5 Individual Model Performance Comparison

The Competition Model (KNN) exhibited the most significant improvement, with a +6.69% increase in ROC-AUC upon excluding COVID-19 data, followed by the Volatility Model, which showed a +2.39% increase. Table 5 suggests that spatial and competitive dynamics, as well as business lifecycle patterns, become more stable, consistent, and easier to model when pandemic-related disruptions are removed from the training dataset.

4.6 Model Alternative Results

Further machine learning techniques were investigated during the optimization phase to enhance predictive performance beyond the baseline weighted fusion approach. In line with insights from the

Table 5: Individual Model Performance - COVID-19 Comparison

Model	Covid Included	Covid Excluded	Improvement
Competition (KNN)	63.11%	69.80%	+6.69%
Volatility (LR)	82.84%	85.23%	+2.39%
Popularity (LR)	58.00%	58.84%	+0.80%

literature review indicating that ensemble methods have demonstrated strong effectiveness in comparable prediction tasks [9]. Among the approaches evaluated found at Table 6, the Voting Ensemble with SMOTE achieved the best results, attaining 85.12% accuracy and 91.18% ROC-AUC, which correspond to improvements of +11.64% in accuracy and +13.29% in ROC-AUC compared to the baseline weighted fusion model.

Table 6: Model Optimization Results

Method	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (Weighted Fusion)	73.48%	88.20%	74.76%	80.93%	77.89%
Random Forest + SMOTE	83.72%	88.72%	84.89%	86.76%	91.40%
XGBoost + SMOTE	83.72%	85.82%	87.05%	86.43%	91.57%
Voting Ensemble + SMOTE	85.12%	88.49%	88.49%	88.49%	91.18%

4.7 Performance Against Benchmark Requirements

The benchmark targets were defined and grounded in existing literature on business viability prediction models. [10] Based on Table 7, reported F1-scores ranging from 76% to 94% for feasibility prediction systems. Guided by this range, the study established two reference benchmarks: an aspirational target of 90% accuracy and a literature-informed average benchmark of 85%.

Table 7: Performance Against Benchmark Requirements

Metric	Target	Baseline	Optimized	Status
Accuracy	90%	73.48%	85.12%	Not Met
F1-Score	85% (avg)	80.93%	88.49%	Met
ROC-AUC	90%	77.89%	91.18%	Exceeds

4.8 Performance Within Literature Benchmarks

Based on Table 8, the model achieved an accuracy of 85.12%, which meets the literature average, while its F1-Score of 88.49% exceeds the average by 3.49 percentage points. Most notably, the ROC-AUC of 91.18% surpasses the 90% aspirational target when interpreted as a discrimination metric, indicating excellent performance in ranking businesses according to their probability of viability.

4.9 Summary of Key Findings

1. The Volatility Model (Logistic Regression) demonstrated the strongest standalone performance, achieving a ROC-AUC of 85.23%.
2. The feature `has_recent_renewal` was identified as the most influential predictor, with a coefficient of +1.198.
3. The baseline weighted fusion model recorded 77.89% ROC-AUC and 88.20% precision.
4. Optimization enhanced overall performance, reaching 85.12% accuracy and 91.18% ROC-AUC.

Table 8: Comparison with Literature Benchmarks

Metric	Literature Range	Lit. Average	This Study	Assessment
Accuracy	76% - 94%	85%	85.12%	Meets Avg
F1-Score	76% - 94%	85%	88.49%	Exceeds Avg
ROC-AUC	–	–	91.18%	Excellent

5. The 90% accuracy benchmark was not attained due to dataset constraints and inherent noise; however, results remain aligned with the literature average (85%), with ROC-AUC exceeding 90%.
6. Restaurants exhibited the highest risk profile (31.3% not feasible), while Hotels/Resorts showed the strongest viability (58.1% feasible).
7. Excluding COVID-19 data improved model discrimination across individual models by approximately 2–7% ROC-AUC.

5 Conclusion

5.1 Summary of the Study

This research developed and validated an integrated SME Business Viability Assessment System for the scope in Bohol province, using a machine learning framework consisting of Logistic Regression for the Popularity and Volatility models, K-Nearest Neighbors (KNN) for the Competition model, and a weighted scoring mechanism for integrating outputs. The system was trained on a dataset of 628 SMEs across four categories sourced from multiple datasets, with a temporal split applied to exclude the COVID-19 period (2020–2021) to ensure cleaner model learning. Model optimization further incorporated ensemble techniques to evaluate performance improvements beyond the baseline methodology.

5.2 Conclusions

5.2.1 Achievement of the Objective of the Study

The study successfully gathered and curated a dataset comprising 628 businesses, from which 25 features were extracted covering popularity, turnover, and competitive density. It implemented Logistic Regression for the Popularity and Volatility components (84.09% and 89.44% F1-scores, respectively) and K-Nearest Neighbors for the Competition component (69.80% ROC-AUC), with a weighted fusion approach achieving 77.89% ROC-AUC. The system was further validated using a temporal split, yielding 88.20% precision under the baseline model and improving to 91.18% ROC-AUC under the optimized configuration, alongside the application of a tri-band classification scheme.

5.2.2 Baseline vs. Optimized Performance

The baseline configuration (Logistic Regression + KNN + Weighted Fusion) achieved 77.89% ROC-AUC and 88.20% precision, while the optimized Voting Ensemble with SMOTE improved performance to 91.18% ROC-AUC and 85.12% accuracy. Despite the performance gain, the Voting Ensemble was not selected as the primary architecture due to reduced interpretability and limited explainability at the component level. Additionally, its lack of support for partial inference when feature groups are unavailable further constrained its suitability for deployment.

5.2.3 Performance Against Benchmark Requirements

The study established an aspirational benchmark of 90% accuracy based on literature standards [10], which was not achieved due to dataset size constraints, unobserved external factors (e.g., management

quality and market timing), limited feature availability (absence of financial variables), and class imbalance. Despite this, the model remains competitive with literature benchmarks, achieving 85.12% accuracy (aligned with the 85% average) and an F1-score of 88.49% (+3.49%). Most notably, the ROC-AUC of 91.18% exceeds the 90% benchmark, indicating strong discriminative capability in correctly ranking viable versus non-viable businesses.

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